

Quoc Cuong LE

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Education

- PhD University of Tours**, Computer Science Tours, France
Nov 2016 – Mar 2020
- Laboratory of Fundamental and Applied Computer Science of Tours
 - Research on multiple object tracking within a distributed and collaborative multiple-camera network.
 - PhD Thesis** [🔗](#)
 - Course** - Computer Vision (Camera Calibration, Object Recognition, Object Tracking, Graph Theory, Graph matching, Graph Segmentation) Machine Learning (Statistical Inference, Bayesian Methods, Optimization theory, Deep Generative models (GANs, VAEs), Reinforcement Learning, Representation Learning)
- MS Institut National des Sciences Appliquées Centre Val de Loire**, Industrial System Engineering Blois, France
Sept 2011 – Sept 2016
- Major in Industrial System Engineering | Minor in Automation System, Industrial informatics, and Instrumentation (GPA 4.0, Rank A)
 - Relevant course** - Advanced Linear Algebra, Advanced Probability and Statistics, Scientific Computing and Numerical Analysis, Advanced Signal Processing.
- MS University of Orléans**, Control Engineering (Dual-degree programme with INSA Centre Val de Loire) Blois, France
Sept 2015 – Sept 2016
- Mechatronics, Control, Robotics, and Signal (GPA 3.9, Rank A)
 - Relevant course** - Numerical filters (.e.g. Kalman Filter), Advanced Control Theory, Optimal Control, Robotics.

Professional Experience

- XXII GROUP**, Research Scientist Puteaux, France
Nov 2022 – May 2024
- Conducted multiple research projects in Computer Vision and Machine Learning
 - Developed a Real-time Multi-Modal Open-Vocabulary/ Zero/One/Few-shot Object Detection (named YOLO-CLIP) based on Vision-Language Pretrained models, e.g. CLIP (OpenAI), GLIP (Microsoft)
 - Trained, scaled, and optimized ML models with distributed Multi-Node Multi-GPU training process.
 - Build a fast and robust detection model training with Self-Supervised Learning (SSL) pre-trained approaches e.g. DINO, MoCo, SimCLR
 - Developed a Group Distribution Robust Optimization approach to Out-of-Distribution and Distribution Shift problems in Object Recognition, Detection, and Segmentation.
 - Participated in external R&D projects in collaboration with multinational teams.
 - Collaborated with European team in a consortium of EU Starlight project (codename H2020) as a partner providing vision-related technologies including Detection, Tracking, and Re-Identification.
 - Tested and benchmarked hardware AI accelerators by Axelera AI for computer vision applications
- XXII GROUP**, Research Engineer Puteaux, France
Apr 2020 – Nov 2022
- Scalable real-time solutions for Smart City, and Retail such as traffic monitoring, incident detection, and other video surveillance applications
 - Developed a fast Multiple Object Tracking algorithm as a common shared library used in all service projects, and our main products XXII-Smartcity, Retail, Infra, and Replay.
 - Developed multi-camera tracking systems for adapted enclosed environments such as retail shops, supermarkets, etc.
 - Leading role in multi-team collaboration projects as an expert in tracking algorithms- Developed a benchmarking system for video surveillance systems.- Developed multi-modal non-biometric re-Identification, image/video retrieval systems.
 - Technology: Python Poetry Pydantic FastAPI Pytorch Tensorflow 2 Tensorstore OpenCV Apache NiFi CI/CD Scrum/Agile
- CEA LIST**, Research Intern Saclay, France
Mar 2016 – Sept 2016
- Implement multiple signal interpolation methods to speed up the Non-Destructive Testing (NDT) simulation of ultrasound echos in CIVA, a simulation and analysis software for NDT.
 - Keyword:** Auto-correlation filter, convolution filter, Auto-Regressive Moving Average filter.

Scholarship & Award

France Excellence Eiffel scholarship [↗](#)

2015 – 2016

Awarded by the French Ministry for Europe and Foreign Affairs to top foreign students in France

Scholarship of Odon Vallet Foundation [↗](#)

2014 – 2015

Third prize in National Contest in Physics for high school students in Vietnam (#108)

2009

Publications & Patents

[Patent] Method for analyzing video content and device executing the method

BADRINATH Anand, DE CASTILLA David, PEREZ Alexandre, **LE Quoc Cuong**, BENTERKIA Amine, DERRAGUI Amine, HANOUNE Souheil, MARCHETTI Antoine

patents.google.com/patent/FR3126166A1/en [↗](#)

Unbalanced Optimal Transport in Multi-Camera Tracking Applications

Mar 2021

Quoc Cuong LE, Donatello CONTE, Moncef HIDANE

[10.1007/978-3-030-68821-9_30](https://doi.org/10.1007/978-3-030-68821-9_30) [↗](#) (International Conference on Pattern Recognition. T. 12665. ICPR 2021 : Pattern Recognition. ICPR International Workshops and Challenges. Milan, Italy: Springer International Publishing, p. 327-343.)

Appearance features for online multiple camera multiple target tracking

Mar 2020

Quoc Cuong LE, Moncef HIDANE

[10.1145/3341105.3373960](https://doi.org/10.1145/3341105.3373960) [↗](#) (SAC'20: 35th Annual ACM Symposium on Applied Computing. Brno, Czech Republic.)

Online Multiple View Tracking: Targets Association Across Cameras

Sept 2018

Quoc Cuong LE, Donatello CONTE, Moncef HIDANE

hal.science/hal-01880374 [↗](#) (6th Workshop on Activity Monitoring by Multiple Distributed Sensing (AMMDS2018). Newcastle, United Kingdom.)

ML & CV Projects

AI Semantic Document Search Tools [↗](#)

June 2024 – Sept 2024

Using the Retrieval Augmented Generation (RAG) technique to find relevant multilingual content based on text embedding, this project provides an alternative search approach to web-based search in complex domains such as Legal and technical/ scientific documents.

- Retrieval algorithms: BM25, rerank, lexical and semantic search
- Local deployment via Ollama or via AI cloud services, e.g. ChatGPT, Gemini, etc.
- **Mini extension**: cover letter generation based on matched information between user portfolio and job description retrieved by RAG

Visual SLAM [↗](#)

Feb 2025 – present

Build a real-time visual SLAM with robust camera pose estimation

- Superpoint for keypoint extraction
- Superglue for point tracking
- Robust PnP algorithms
- **Keywords**: Camera Calibration, Online Calibration, State Estimation, Sensor Fusion, 3D Reconstruction, Camera Pose Estimation, Structure-From-Motion

Skills

Programming: Proficient with Python, C++, and Git; good understanding of Web, app development, and DevOps

ML Framework: Pytorch, Tensorflow 2, JAX, MLflow, Open MMLab, Google's frameworks (Tensorflow Model Garden, Scenic), TensorRT

Mathematics: Good understanding of differential equations, calculus, and linear algebra

Languages: English (fluent, C1 Cambridge English Certificate (07/2022), TOEIC 975/990), French (fluent), Vietnamese (native)